h1 — First technical assessment

> **Enquiry:** Our existing NIR pilot follows API concentration in one powder formulation at the tablet press feed frame. We are introducing a different API and excipient blend and would like to reuse the calibration. A colleague suggests adding Raman so that the old calibration will remain valid. What should our first technical assessment and experiment be?
>
> **Answers added:** Path B - Where it is used: In production; Path C - Variables per sensor: One

Quality checks: 15/16 · **Sourced claims verified:** 6/8 · **Unsupported numbers:** 0

01. Decision Status

Provisional assessment · **Confidence:** low (computed, not self-reported)

The existing NIR calibration cannot be directly reused for the new blend; a re-validation of NIR and a feasibility test of a fused NIR-Raman chemometric model are required.

The customer's own claims, tested first

- **unverifiable**: "Adding Raman will keep the old calibration valid". Adding a second method (for example Raman) does not make an existing single-method calibration valid. Data from different methods are not equivalent; a fused model needs synchronised data, fusion algorithms and its own calibration and validation. [S01 p.8] [S01 p.9]
- **unverifiable**: "Reuse existing NIR calibration for new API/excipient blend". NIR and other spectra need chemometric calibration models built and validated on the actual matrix. A new API or excipient blend changes the spectra, so an existing calibration must be re-validated and normally rebuilt; reuse is a hypothesis to test, not an assumption. [S01 p.10] [S01 p.5]

To raise confidence, answer: Is the sample clear, turbid or opaque, and does this change during the process? / What concentration range must be covered, and what is the smallest change that matters? / Which independent reference method (and its uncertainty) can provide calibration values?

02. Requirements stated

- Assess Raman’s impact on NIR calibration for new powder blend
- Test combined spectra and chemometric model transfer
- Target variable: API concentration in powder at tablet press feed frame
- Phase: pilot (powder)
- Existing asset: NIR pilot

03. Key unknowns

- Exact matrix composition of the new API/excipient blend
- Optical scattering and absorption behavior of the new blend in the NIR and Raman regions
- Concentration range of the API in the new blend

Assumptions made

- The new API/excipient blend remains a free-flowing powder suitable for diffuse-reflection NIR probing.
- A Raman probe with 785 nm excitation can be installed without major mechanical redesign.

04. Clarification questions

1. **What is the expected API concentration range in the new blend?** *(high impact)*. Why: Defines the calibration window and influences the number of calibration samples needed.

2. **Can you provide representative powder samples covering the full range of excipient variations you anticipate?** *(high impact)*. Why: Matrix variability drives the need for a robust chemometric model and affects transferability of the existing NIR calibration.
3. **What Raman excitation wavelength and probe geometry are available in your plant?** *(medium impact)*. Why: Raman signal strength and fluorescence background depend on excitation wavelength; the probe must be compatible with the feed-frame environment.
4. **Do you have an independent reference assay (e.g., HPLC) for API concentration on the same samples?** *(medium impact)*. Why: Reference data are required for model validation and to quantify any improvement from adding Raman.
5. **Is the existing NIR probe fixed in the feed frame, and can a Raman probe be co-located on the same shaft?** *(low impact)*. Why: Physical integration determines whether low-level data fusion (synchronized spectra) is feasible.

05. Recommendation

NIR (diffuse reflection) – re-validation (primary)

- NIR is the proven method for powder blends in the tablet press feed frame and already deployed in the pilot. [S01 p.5] *(sourced)*
- Re-validate the existing NIR calibration on the new blend using a representative calibration set and independent reference assays. *(proposal)*
- *Applies if:* Only if a sufficient set of new-blend samples covering the client-agreed concentration range is available.

Raman (785 nm contact probe) – feasibility test (conditional)

- Raman provides complementary chemical information and is supported for powder blends in the feed frame. [S01 p.5] *(sourced)*
- Test Raman alone to assess signal quality, fluorescence background, and linearity for the new API. *(proposal)*
- *Applies if:* Requires a Raman probe compatible with the existing shaft and a fluorescence assessment.

Fused NIR-Raman (low-level data fusion) – experimental comparison (comparison)

- Published work shows fused NIR-Raman models can improve accuracy and robustness for API potency in the feed frame, but this is not a guarantee for a new formulation. [S01 p.15] *(sourced)*
- Build and compare three models (NIR only, Raman only, fused) on the same calibration set to quantify any benefit. *(proposal)*
- *Applies if:* Requires synchronized acquisition of NIR and Raman spectra on each sample.

Method screening (deterministic, from sourced rules)

Method	Score	Verdict
NIR spectroscopy	3	lead
Raman spectroscopy	2	candidate
UV/Vis spectroscopy	0	weak
Fluorescence spectroscopy	0	weak
MIR spectroscopy (ATR)	-1	weak

06. Measurement concept

- Collect synchronized NIR diffuse-reflection spectra and Raman spectra on representative powder samples, then build chemometric models (PLS) for API concentration using NIR alone, Raman alone, and fused NIR-Raman data. *(proposal)*
- Validate each model against independent reference assays to quantify accuracy, robustness, and any benefit of the fused approach. *(proposal)*

07. Spectral range

- NIR diffuse-reflection: client-agreed range (broad overlapping bands). [S01 p.5] *(sourced)*
- Raman: Raman shift range typically client-agreed; excitation wavelength 785 nm (as used in pharmaceutical feed-frame probes). [S01 p.15] *(sourced)*

08. Risks and uncertainty

- Fluorescence background may mask Raman signals in organic powders. [S01 p.5] *(sourced)*
- Matrix change may render the existing NIR calibration inaccurate. [S01 p.5] *(sourced ⚠ unverified)*
- Adding Raman does not automatically preserve the validity of the original NIR model. [S01 p.5] *(sourced ⚠ unverified)*

Confidence: low

- Information: low (2/10 decision-relevant facts known; target defined: yes)
- Evidence: high (Direct application in AFS sources: R_PHARMA_BLEND)
- Scope fit: high (Requested use matches the scope of the cited evidence)
- Client premises: medium (0 contradicted, 2 unverifiable, 0 supported)
- Verification: medium (6/8 sourced claims verified; 0 unsupported numbers)
- Model agreement: high (3/3 drafts share the lead method(s) NIR; 3/3 share the status 'provisional_assessment' (2 model families))

09. Decision basis

- Client-agreed accuracy and robustness criteria on independent reference data
- Demonstrated improvement (or lack thereof) of fused model over NIR-only model

10. Sources

- **S01 p.5**: Multispectral Laboratory of Optical Analysis Methods: purpose, methods, and applications (art photonics / MIR Photonics) *(overview)*
  - "NIR spectroscopy, with its relatively large penetration depth (up to several millimetres) even into dense solid samples due to low absorption and strong scattering, allows the use of ordinary glass for optics"
  - "the spectra of vibrational transitions in Raman and mid-IR spectroscopy carry complementary qualitative and quantitative information about the sample's composition."
  - "However, most chemicals do not exhibit this property."
  - "NIR and other spectra need chemometric calibration models built and validated on the actual matrix. A new API or excipient blend changes the spectra, so an existing calibration must be re-validated and normally rebuilt" ⚠ unverified
- **S01 p.8**: Multispectral Laboratory of Optical Analysis Methods: purpose, methods, and applications (art photonics / MIR Photonics) *(overview)*

    - "It is crucial that the spectral information obtained by different methods—Mid-IR, NIR, Raman, and fluorescence spectra—is collected from the same small, even pinpoint, area of the sample."
- **S01 p.9**: Multispectral Laboratory of Optical Analysis Methods: purpose, methods, and applications (art photonics / MIR Photonics) *(overview)*
    - "The development of OMS begins with in-depth studies of the analyzed object using standard laboratory spectrometers."
- **S01 p.10**: Multispectral Laboratory of Optical Analysis Methods: purpose, methods, and applications (art photonics / MIR Photonics) *(overview)*
    - "These immersion probes operate on the ATR principle but allow analysis at a significant distance from the spectrometer, without the need for sampling."
- **S01 p.15**: Multispectral Laboratory of Optical Analysis Methods: purpose, methods, and applications (art photonics / MIR Photonics) *(overview)*
    - "By combining Raman and NIR, these models can predict API potency or coating thickness with higher accuracy and robustness than single-method models."
    - "Raman spectrometer and Raman fiber probe for contact measurements with 785 nm excitation wavelength"

Verifier warnings

- risks[1]: unverified S01 p.5: quote not on page (49)
- risks[2]: unverified S01 p.5: quote not on page (49)

11. First feasibility experiment

Samples

- Representative powder batches covering the full API concentration range and expected excipient variations **Reference measurements**
- Independent HPLC assay for API concentration on each calibration and validation sample **Comparison**
- Model performance (RMSEP, bias) for NIR-only, Raman-only, and fused NIR-Raman models **Steps**

1. Prepare a calibration set (≈30–50 samples) spanning the API concentration range and matrix variability.
2. Acquire synchronized NIR diffuse-reflection spectra and Raman spectra (785 nm) for each sample using the existing NIR probe and a compatible Raman probe.
3. Perform fluorescence/background assessment on Raman spectra; adjust acquisition parameters if needed.
4. Build three chemometric PLS models (NIR, Raman, fused) using standard preprocessing (scatter correction, baseline removal).
5. Validate each model against the independent HPLC reference set; compute RMSEP, bias, and robustness metrics.
6. Review results with the client; decide whether to adopt the existing NIR model, switch to a new NIR model, or implement a fused NIR-Raman solution.

Next step with AFS

- Conduct a multi-method lab trial (NIR + Raman) to generate calibration data and evaluate fused-model performance before any production deployment.
- Likely path: A - Path A aligns with the goal of real-time concentration monitoring for process automation; the lab trial provides the evidence needed to select the optimal model and hardware configuration.
- Path A (chosen by the expert): **fits**. Path A turns spectral data into real-time concentration values for process automation; a central full-spectrum analyzer gives the higher accuracy. [S02 p.20, S02 p.18]

- Path B (chosen by the expert): **conflicts**. Path B is a lab system for R&D; measuring in production is Path A (central analyzer) or Path C (sensors). [S02 p.20]
- Path C (chosen by the expert): **open**. No finding speaks for or against this path yet; the lab trial decides. [S02 p.20]

If you proceed anyway with Path B (Path B (customised multi-spectral system for R&D) is a poor fit here: Path B is a lab system for R&D; measuring in production is Path A (central analyzer) or Path C (sensors).)

1. Specify the lab kit from the methods short-listed in this assessment: spectrometers, fiber probes, software.
2. Run the method comparison on the customer's own samples in their lab.
3. Use the results as the basis for a later process solution (Path A or C). *Sources: S02 p.20, S01 p.4*

Proposed setup (ranked by recommendation; validate in a lab trial)

System	Path A: Chemometric model for process automation	Path B: Multi-spectral system for your R&D lab	Path C: Low-cost spectral sensors
1. NIR (diffuse reflection) – re-validation (primary)	Models: API concentration calibration → Process predictions	Lab system: NIR measurement channel, Sample / probe, Acquisition + reference data → Spectra + reference values	Sensor system: NIR sensor, Process probe(s), Acquisition + control interface → Process monitoring
2. Fused NIR-Raman (low-level data fusion) – experimental comparison (comparison)	Models: API concentration calibration → Process predictions	Lab system: NIR measurement channel, RAMAN measurement channel, Sample / probe, Acquisition + reference data → Spectra + reference values	Sensor system: NIR sensor, RAMAN sensor, Process probe(s), Acquisition + control interface → Process monitoring
3. Raman (785 nm contact probe) – feasibility test (conditional)	Models: API concentration calibration → Process predictions	Lab system: RAMAN measurement channel, Sample / probe, Acquisition + reference data → Spectra + reference values	Sensor system: RAMAN sensor, Process probe(s), Acquisition + control interface → Process monitoring

Council

Winner: draft A; agreement 100% (3/3 drafts share the lead method(s) NIR; 3/3 share the status 'provisional_assessment')

- Draft A (openai/gpt-oss-120b): judge score 5.0, gate failures: G1 Sourced claims verified on the cited page, G2 No unsupported numbers
- Draft B (qwen/qwen3.8-27b): judge score 5.0, gate failures: G1 Sourced claims verified on the cited page, G2 No unsupported numbers
- Draft C (openai/gpt-oss-20b): judge score 4.0, gate failures: G2 No unsupported numbers

Generated by the AFS assessment assistant (run 6909d89206, mode council, server default models; models used: openai/gpt-oss-120b, openai/gpt-oss-20b, openai/gpt-oss-20b -> rule fallback, qwen/qwen3.8-27b).

h2 — First technical assessment

> **Enquiry:** Our dairy process handles both a relatively clear whey stream and an opaque milk concentrate. We want a pilot for composition monitoring, using laboratory protein and fat results where those variables are relevant. We can compare different measurement interfaces before choosing the installation. What first measurement concepts and comparison experiment would make sense for these two streams?
>
> **Answers added:** Path B - Where it is used: In production; Path C - Variables per sensor: Two or three

Quality checks: 12/16 · **Sourced claims verified:** 9/10 · **Unsupported numbers:** 1

01. Decision Status

Provisional assessment · **Confidence:** insufficient (computed, not self-reported)

A pilot using NIR reflection for the opaque milk concentrate and NIR transmission/transflection for the clear whey stream, with optional MIR-ATR or Raman probes as comparative alternatives, is the most straightforward first-step approach.

Problems found in the request

- Fluorescence needs a fluorescent target: Fluorescence has high sensitivity but low selectivity, and most chemicals do not fluoresce. Without a defined fluorescent target, marker or reference relationship, sensitivity alone does not justify reporting a concentration. [S01 p.5, S01 p.6, S04 p.4]

The customer's own claims, tested first

- **supported**: "We want a pilot for composition monitoring, using laboratory protein and fat results where those variables are relevant.". NIR reflection is the standard for opaque or concentrated dairy streams and transmission for clear streams, with scatter correction and PLS for protein and fat quantification. The pilot can use lab protein/fat data to build calibration models. [S01 p.17] [S01 p.18] [S01 p.18]
- **supported**: "We can compare different measurement interfaces before choosing the installation.". Rule R_CERTAINTY_DEMAND states that method selection depends on target, sample optics, and acceptance criteria, and must be decided by comparison on real samples. Therefore a comparison experiment is appropriate. [S01 p.5] [S01 p.18]

To raise confidence, answer: What is the sample (liquid, powder, melt, tissue...) and its main components? / Is the sample clear, turbid or opaque, and does this change during the process? / What concentration range must be covered, and what is the smallest change that matters?

02. Requirements stated

- protein
- fat

03. Key unknowns

- exact composition ranges of whey and milk concentrate
- temperature of streams
- fiber cable type and length

Assumptions made

- The whey stream is clear enough for transmission or transflection; the milk concentrate is opaque; fiber lengths are within the attenuation limits of the chosen methods.

04. Clarification questions

1. **What are the expected protein and fat concentration ranges for the whey and milk concentrate streams?** *(high impact)*. Why: Calibration ranges and model robustness depend on the full concentration span.
2. **What is the operating temperature of the streams during measurement?** *(medium impact)*. Why: Temperature influences spectral baseline and scattering.
3. **What fiber cable lengths and types (silica, chalcogenide, etc.) are planned for the pilot?** *(medium impact)*. Why: Fiber attenuation and penetration depth affect signal quality.
4. **Will the measurement be performed inline in the production line or in a lab-based loop?** *(low impact)*. Why: It determines the choice of probe geometry and installation constraints.

05. Recommendation

NIR (reflection) (primary)

- NIR reflection is the workhorse for opaque or concentrated dairy products and is used for fat and protein quantification with scatter correction and PLS. [S01 p.5] *(sourced)*
- Build a calibration model using lab protein/fat reference data for the milk concentrate. *(proposal)*
- *Applies if:* Representative milk concentrate samples covering the full protein/fat range must be available.

NIR (transmission/transflection) (primary)

- Clear or semi-transparent liquids are best measured with transmission or transflection probes with a defined 1-10 mm path. [S01 p.5] *(sourced)*
- Use a transflection probe that can also serve as a reflection probe for the milk concentrate to simplify installation. *(proposal)*
- *Applies if:* The whey stream must be clear enough for transmission; a 1-10 mm path length should be verified.

MIR (ATR) (comparison)

- MIR ATR provides specific absorption bands for fat and protein and can be used for aqueous media. [S01 p.5] *(sourced)*
- Test an ATR probe on both streams to evaluate specificity versus NIR. *(proposal)*
- *Applies if:* ATR probe must be compatible with the stream temperature and flow.

Raman (comparison)

- Raman signals are weak but can be useful for aqueous media and are not strongly absorbed by water. [S01 p.5] *(sourced)*
- Assess Raman background and signal quality on the whey and milk concentrate. *(proposal)*
- *Applies if:* A suitable laser wavelength and probe design must be selected to avoid fluorescence.

Fluorescence (not recommended)

- Fluorescence requires a fluorescent target; most dairy components do not fluoresce, making it unsuitable for protein/fat quantification. [S01 p.5] *(sourced)*
- *Applies if:* None; not recommended for this application.

Method screening (deterministic, from sourced rules)

Method	Score	Verdict
NIR spectroscopy	5	lead
MIR spectroscopy (ATR)	1	candidate
Raman spectroscopy	1	candidate
UV/Vis spectroscopy	0	weak
Fluorescence spectroscopy	-3	not supported

06. Measurement concept

- The pilot will use laboratory protein and fat results as reference for calibration. *(client fact)*
- NIR reflection is standard for opaque dairy streams and transmission for clear streams. [S01 p.5] *(sourced)*
- MIR ATR and Raman can add specificity but require careful probe selection. [S01 p.5] *(sourced)*

07. Spectral range

- NIR: client-agreed range for protein/fat quantification. [S01 p.5] *(sourced)*
- MIR ATR: 4000–400 cm^{-1}, focusing on 1740 cm^{-1} (fat) and 1650 cm^{-1} (protein). [S04 p.3] *(sourced)*
- Raman: 400–4000 cm^{-1} with excitation at 785 nm (typical for dairy). [S01 p.11] *(sourced ⚠ unverified)*

08. Risks and uncertainty

- The whey stream will remain clear enough for transmission; if it becomes turbid, the NIR transmission probe may fail. *(assumption)*
- Temperature variations may shift spectral baselines; calibration models must account for this. *(assumption)*
- Fiber attenuation over long lengths could reduce signal strength, especially for MIR ATR. *(assumption)*
- Include temperature sensors and monitor optical path length during the pilot. *(proposal)*

Confidence: insufficient

- Information: insufficient (1/10 decision-relevant facts known; target defined: yes)
- Evidence: high (Direct application in AFS sources: R_DAIRY)
- Scope fit: high (Requested use matches the scope of the cited evidence)
- Client premises: high (0 contradicted, 0 unverifiable, 2 supported)
- Verification: high (9/10 sourced claims verified; 1 unsupported numbers)
- Model agreement: high (3/3 drafts share the lead method(s) NIR; 3/3 share the status 'provisional_assessment' (2 model families))
- Cap: R_FLUO_NEEDS_TARGET: cap low
- Cap: R_CERTAINTY_DEMAND: cap low

09. Decision basis

- Model R^2 and RMSEP against lab reference; acceptance criteria set by client (e.g., RMSEP < client-agreed value)

10. Sources

- **S01 p.5**: Multispectral Laboratory of Optical Analysis Methods: purpose, methods, and applications (art photonics / MIR Photonics) *(overview)*
  - "NIR spectroscopy, with its relatively large penetration depth (up to several millimetres) even into dense solid samples due to low absorption and strong scattering, allows the use of ordinary glass for optics"
  - "strong absorption in the mid-IR region limits the corresponding spectral method to the analysis of very thin samples (usually less than 50 µm), but it opens up the possibility of using attenuated total reflectance (ATR)"
  - "Moreover, the spectra of vibrational transitions in Raman and mid-IR spectroscopy carry complementary qualitative and quantitative information about the sample's composition."
  - "However, most chemicals do not exhibit this property."
  - "selecting a spectral method for a specific analytical problem is a complex task requiring specialized knowledge in chemistry, physics, and mathematics."
- **S01 p.11**: Multispectral Laboratory of Optical Analysis Methods: purpose, methods, and applications (art photonics / MIR Photonics) *(overview)*
  - "Raman signals are weak and, for many organic or biological samples, fluorescence can overlay them." ⚠ unverified
- **S01 p.17**: Multispectral Laboratory of Optical Analysis Methods: purpose, methods, and applications (art photonics / MIR Photonics) *(overview)*
  - "Inline NIR Reflection is primarily used for opaque or highly concentrated dairy products"
- **S01 p.18**: Multispectral Laboratory of Optical Analysis Methods: purpose, methods, and applications (art photonics / MIR Photonics) *(overview)*
  - "NIR Spectrometer and NIR light source remain the same for both reflection and transmission techniques"
  - "advanced labs often incorporate Fluorescence, Raman, and Mid-Infrared (MIR) ATR fiber optic techniques to capture specific chemical or structural information that NIR might miss"
- **S04 p.3**: Fiber Probe for Simultaneous Mid-Infrared and Fluorescence Spectroscopic Analysis (Anal. Chem. 2021) *(lab_demonstration)*
  - "the stretching vibration band of the ester carbonyl C=O at about 1740 cm−1, which is characteristic of saturated triglycerides"

Verifier warnings

- spectral_range[2]: unverified S01 p.11: quote not on page (53)

11. First feasibility experiment

Samples

- Representative whey samples covering the full protein/fat range
- Representative milk concentrate samples covering the full protein/fat range **Reference measurements**
- Independent lab assays of protein (Kjeldahl) and fat (Gerber) for each sample **Comparison**
- NIR reflection vs NIR transmission/transflection vs MIR ATR vs Raman **Steps**

1. Install NIR reflection probe on milk concentrate line and NIR transmission/transflection probe on whey line; optionally add MIR ATR and Raman probes on both streams.
2. Collect spectra and reference data for at least 30 samples per stream.

3. Preprocess spectra (MSC/SNV for NIR, baseline correction for Raman).
4. Build PLS models for protein and fat for each method and compare performance.
5. Evaluate robustness to temperature and flow variations by spiking samples and repeating measurements.
6. Select the best performing method(s) for pilot deployment and plan for full-scale installation.

Next step with AFS

- Conduct a multi-method lab trial to build and validate chemometric models for protein and fat on both streams.
- Likely path: A - Path A provides a central full- spectrum analyzer that delivers the highest accuracy for process automation.
- Path A (chosen by the expert): **fits**. Path A turns spectral data into real-time concentration values for process automation; a central full-spectrum analyzer gives the higher accuracy. [S02 p.20, S02 p.18]
- Path B (chosen by the expert): **conflicts**. Path B is a lab system for R&D; measuring in production is Path A (central analyzer) or Path C (sensors). [S02 p.20]
- Path C (chosen by the expert): **conflicts**. Several variables at once need full-spectrum information; a few-wavelength sensor is designed for one specific control task. A full-spectrum analyzer gives higher accuracy. [S02 p.18, S02 p.20]

If you proceed anyway with Path B (Path B (customised multi-spectral system for R&D) is a poor fit here: Path B is a lab system for R&D; measuring in production is Path A (central analyzer) or Path C (sensors).)

1. Specify the lab kit from the methods short-listed in this assessment: spectrometers, fiber probes, software.
2. Run the method comparison on the customer's own samples in their lab.
3. Use the results as the basis for a later process solution (Path A or C). *Sources: S02 p.20, S01 p.4*

If you proceed anyway with Path C (Path C (customised spectral IoT sensor) is a poor fit here: Several variables at once need full-spectrum information; a few-wavelength sensor is designed for one specific control task. A full-spectrum analyzer gives higher accuracy.)

1. Run a lab trial with full-spectrum instruments on representative samples.
2. Select the most informative wavelengths and check that a few-band model reaches the client-agreed error on independent samples.
3. Build and test a prototype LED / filter sensor against the reference method before any volume order. *Sources: S01 p.9, S03 p.14, S02 p.18*

Proposed setup (ranked by recommendation; validate in a lab trial)

System	Path A: Chemometric model for process automation	Path B: Multi-spectral system for your R&D lab	Path C: Low-cost spectral sensors
1. NIR (reflection) (primary)	Models: protein calibration, fat calibration → Process predictions	Lab system: NIR measurement channel, Sample / probe, Acquisition + reference data → Spectra + reference values	Sensor system: NIR sensor, Process probe(s), Acquisition + control interface → Process monitoring
2. NIR (transmission/transflection) (primary)	Models: protein calibration, fat calibration → Process predictions	Lab system: NIR measurement channel, Sample / probe, Acquisition + reference data → Spectra + reference values	Sensor system: NIR sensor, Process probe(s), Acquisition + control interface → Process monitoring

System	Path A: Chemometric model for process automation	Path B: Multi-spectral system for your R&D lab	Path C: Low-cost spectral sensors
3. MIR (ATR) (comparison)	Models: protein calibration, fat calibration → Process predictions	Lab system: MIR measurement channel, Sample / probe, Acquisition + reference data → Spectra + reference values	Sensor system: MIR sensor, Process probe(s), Acquisition + control interface → Process monitoring
4. Raman (comparison)	Models: protein calibration, fat calibration → Process predictions	Lab system: RAMAN measurement channel, Sample / probe, Acquisition + reference data → Spectra + reference values	Sensor system: RAMAN sensor, Process probe(s), Acquisition + control interface → Process monitoring

Council

Winner: draft C; agreement 100% (3/3 drafts share the lead method(s) NIR; 3/3 share the status 'provisional_assessment')

- Draft A (openai/gpt-oss-120b): judge score 4.08, gate failures: G1 Sourced claims verified on the cited page, G2 No unsupported numbers
- Draft B (qwen/qwen3.8-27b): judge score 4.83, gate failures: G1 Sourced claims verified on the cited page, G2 No unsupported numbers
- Draft C (openai/gpt-oss-20b): judge score 4.92, gate failures: G1 Sourced claims verified on the cited page, G2 No unsupported numbers

Generated by the AFS assessment assistant (run 4f27f123b4, mode council, server default models; models used: openai/gpt-oss-120b, openai/gpt-oss-20b, qwen/qwen3.8-27b).

h3 — First technical assessment

> **Enquiry:** We already have a fluorescence spectrometer and would like to use it to report glucose concentration and viable biomass directly during fermentation. Our only justification is that fluorescence can be very sensitive. We have not identified a fluorescent target, marker or reference relationship. What should the assistant tell us?
>
> **Answers added:** Path B - Where it is used: Our R&D lab; Path C - Variables per sensor: Two or three

Quality checks: 15/16 · **Sourced claims verified:** 5/5 · **Unsupported numbers:** 0

01. Decision Status

Provisional assessment · **Confidence:** low (computed, not self-reported)

Fluorescence alone is not supported for direct glucose or viable biomass measurement; a multimodal approach with Raman and UV/Vis is recommended for testing.

The customer's own claims, tested first

- **unverifiable**: "fluorescence can be very sensitive". Turbidity / optical density correlates with cell density but does not by itself distinguish viable from non-viable cells. A viable-biomass claim needs a defined indicator and a viability reference method. [S01 p.16] [S01 p.17]
- **unverifiable**: "reuse existing fluorescence spectrometer for glucose and biomass measurement". NIR and other spectra need chemometric calibration models built and validated on the actual matrix. A new API or excipient blend changes the spectra, so an existing calibration must be re-validated and normally rebuilt; reuse is a hypothesis to test, not an assumption. [S01 p.10] [S01 p.5]

To raise confidence, answer: Is the sample clear, turbid or opaque, and does this change during the process? / What concentration range must be covered, and what is the smallest change that matters? / Which independent reference method (and its uncertainty) can provide calibration values?

02. Requirements stated

- glucose concentration
- viable biomass
- fluorescence spectrometer
- fermentation process
- R&D lab

03. Key unknowns

- Fluorescent target or marker for glucose/biomass
- Calibration relationship between fluorescence signal and glucose/biomass concentration
- Matrix composition and optical properties of fermentation broth
- Optical behavior (scattering, absorption) of fermentation broth

Assumptions made

- Assumption: The client's 'fluorescence spectrometer' is a standard benchtop or fiber-coupled instrument capable of measuring emission spectra, not just single-wavelength intensity.
- Assumption: The fermentation broth is optically clear enough for transmission or reflection measurements, or the client is prepared to use ATR or diffuse reflectance geometries if scattering is high.

04. Clarification questions

1. **What specific fluorescent species (e.g., NADH, tryptophan, riboflavin) do you intend to monitor, and do you have a reference method (e.g., HPLC, plate count) to correlate their fluorescence with glucose or biomass?** *(high impact)*. Why: Fluorescence is specific to certain molecules; without a defined target and reference, the signal cannot be interpreted as glucose or biomass concentration.
2. **Have you characterized the optical properties of your fermentation broth (turbidity, background fluorescence, scattering) at the wavelengths of interest?** *(high impact)*. Why: Turbidity and background fluorescence can obscure the signal from the target analyte, making direct measurement impossible without correction or a different method.
3. **What is the expected concentration range of glucose and biomass in your process, and what is the required accuracy/precision?** *(medium impact)*. Why: Fluorescence is sensitive but may lack the dynamic range or specificity for glucose (a non-fluorescent metabolite) and total biomass (a complex mixture) without chemometric modeling.
4. **Are you open to using a combination of methods (e.g., Raman for glucose, UV/Vis for biomass trend, fluorescence for cofactors) rather than relying solely on fluorescence?** *(medium impact)*. Why: No single spectral method is optimal for all bioprocess analytes; a multimodal approach is often more robust and accurate.

05. Recommendation

Raman (primary)

- Raman spectroscopy is well-suited for direct detection of glucose in aqueous fermentation media. [S01 p.5] *(sourced)*
- Raman is particularly well-suited for aqueous environments due to low water absorbance. [S01 p.5] *(sourced)*
- A Raman probe should be tested on the client's fermentation broth to evaluate signal quality and background fluorescence interference. *(proposal)*
- *Applies if:* Only if representative fermentation broth samples with reference assays for glucose and biomass are available.

Fluorescence (conditional)

- Fluorescence spectroscopy is suitable for high-sensitivity detection of specific fluorescent cofactors. [S01 p.5] *(sourced)*
- Most chemicals, including glucose, do not fluoresce, so fluorescence cannot be used directly for glucose measurement. [S01 p.5] *(sourced)*
- The client's existing fluorescence spectrometer can be used to monitor fluorescent cofactors, but a calibration model must be built and validated on the client's specific fermentation broth. *(proposal)*
- *Applies if:* Only if the client is interested in monitoring specific fluorescent cofactors (e.g., NADH, tryptophan) as indicators of metabolic state.

UV/Vis (comparison)

- UV/Vis spectroscopy can be used for real-time biomass monitoring based on turbidity. [S01 p.5] *(sourced)*
- UV/Vis should be tested in parallel with fluorescence and Raman to provide a biomass trend that can be correlated with viable biomass reference measurements. *(proposal)*
- *Applies if:* Only if the client is interested in a low-cost, robust tool for real-time biomass trend monitoring.

Method screening (deterministic, from sourced rules)

Method	Score	Verdict
Raman spectroscopy	4	lead
NIR spectroscopy	2	candidate
MIR spectroscopy (ATR)	1	candidate
UV/Vis spectroscopy	1	candidate
Fluorescence spectroscopy	1	candidate

06. Measurement concept

- Fluorescence spectroscopy is suitable for high-sensitivity detection of specific fluorescent cofactors like NADH, tryptophan, and riboflavin, but not for non-fluorescent metabolites like glucose. *(proposal)*
- Raman spectroscopy provides a molecular fingerprint for direct detection of specific metabolites like glucose, acetate, or lactate in aqueous environments. *(proposal)*
- UV/Vis spectroscopy can be used for real-time biomass monitoring because cell suspension turbidity correlates with cell density, but it does not distinguish viable from non-viable cells. *(proposal)*
- Test a multimodal approach combining Raman (for glucose), UV/Vis (for biomass trend), and fluorescence (for cofactors) to cover the full range of analytes and process variables. *(proposal)*

07. Spectral range

- Raman: Raman shift in cm-1, excitation laser in nm (e.g., 785 nm). *(proposal)*
- Fluorescence: Emission wavelength in nm, excitation wavelength in nm. *(proposal)*
- UV/Vis: Wavelength in nm (e.g., 600 nm for cell density). *(proposal)*

08. Risks and uncertainty

- Turbidity and background fluorescence in the fermentation broth may obscure the signal from the target analyte, making direct measurement impossible without correction or a different method. *(assumption)*
- Fluorescence is specific to certain molecules; without a defined target and reference, the signal cannot be interpreted as glucose or biomass concentration. *(assumption)*
- No single spectral method is optimal for all bioprocess analytes; a multimodal approach is often more robust and accurate. *(assumption)*

Confidence: low

- Information: low (2/10 decision-relevant facts known; target defined: yes)
- Evidence: high (Direct application in AFS sources: R_GLUCOSE_NOT_FLUORESCENT_TARGET, R_BIOPROCESS)
- Scope fit: high (Requested use matches the scope of the cited evidence)
- Client premises: medium (0 contradicted, 2 unverifiable, 0 supported)
- Verification: high (5/5 sourced claims verified; 0 unsupported numbers)
- Model agreement: medium (2/3 drafts share the lead method(s) RAMAN; 3/3 share the status 'provisional_assessment' (2 model families))

09. Decision basis

- Client-agreed accuracy/precision on independent validation samples

- Correlation coefficient (R^2) between spectral signal and reference method (e.g., HPLC for glucose, plate count for biomass)
- Robustness to matrix variation (turbidity, pH, temperature)

10. Sources

- **S01 p.5**: Multispectral Laboratory of Optical Analysis Methods: purpose, methods, and applications (art photonics / MIR Photonics) *(overview)*
  - "the spectra of vibrational transitions in Raman and mid-IR spectroscopy carry complementary qualitative and quantitative information about the sample's composition."
  - "However, most chemicals do not exhibit this property."
  - "the strong overlap of spectral signals in UV spectra complicates the analysis of real samples and makes the use of chemometric algorithms a necessary component"
  - "NIR spectroscopy, with its relatively large penetration depth (up to several millimetres) even into dense solid samples due to low absorption and strong scattering, allows the use of ordinary glass for optics"
- **S01 p.10**: Multispectral Laboratory of Optical Analysis Methods: purpose, methods, and applications (art photonics / MIR Photonics) *(overview)*
  - "These immersion probes operate on the ATR principle but allow analysis at a significant distance from the spectrometer, without the need for sampling."
- **S01 p.16**: Multispectral Laboratory of Optical Analysis Methods: purpose, methods, and applications (art photonics / MIR Photonics) *(overview)*
  - "where no single sensor is universally optimal for every analyte or condition."
- **S01 p.17**: Multispectral Laboratory of Optical Analysis Methods: purpose, methods, and applications (art photonics / MIR Photonics) *(overview)*
  - "Fig. 7. Spectral range for bioprocess monitoring with accessible variables."

11. First feasibility experiment

Samples

- Representative fermentation broth samples across the expected concentration range of glucose and biomass
- Samples with varying turbidity and background fluorescence to test robustness
- Blank samples (media without cells) to characterize background signal **Reference measurements**
- HPLC or enzymatic assay for glucose concentration
- Plate count or dry weight for viable biomass
- UV/Vis absorbance at 600 nm for cell density trend **Comparison**
- Fluorescence alone vs. Fluorescence + Raman vs. Fluorescence + UV/Vis
- Single-wavelength fluorescence vs. full-spectrum fluorescence with chemometrics
- Direct fluorescence measurement vs. fluorescence corrected for turbidity/background

Next step with AFS

- Multi-method lab trial comparing fluorescence, Raman, and UV/Vis on representative fermentation broth samples.
- Likely path: B - Path B gives a tailored set of spectrometers for R&D in the customer's own lab; testing several methods is what research needs.

- Path A (chosen by the expert): **open**. No finding speaks for or against this path yet; the lab trial decides. [S02 p.20]
- Path B (chosen by the expert): **fits**. Path B gives a tailored set of spectrometers for R&D in the customer's own lab. [S02 p.20]
- Path C (chosen by the expert): **conflicts**. Several variables at once need full-spectrum information; a few-wavelength sensor is designed for one specific control task. A full-spectrum analyzer gives higher accuracy. [S02 p.18, S02 p.20]

If you proceed anyway with Path C (Path C (customised spectral IoT sensor) is a poor fit here: Several variables at once need full-spectrum information; a few-wavelength sensor is designed for one specific control task. A full-spectrum analyzer gives higher accuracy.)

1. Run a lab trial with full-spectrum instruments on representative samples.
2. Select the most informative wavelengths and check that a few-band model reaches the client-agreed error on independent samples.
3. Build and test a prototype LED / filter sensor against the reference method before any volume order. *Sources: S01 p.9, S03 p.14, S02 p.18*

Proposed setup (ranked by recommendation; validate in a lab trial)

System	Path A: Chemometric model for process automation	Path B: Multi-spectral system for your R&D lab	Path C: Low-cost spectral sensors
1. Raman (primary)	Models: glucose concentration calibration, viable biomass calibration → Process predictions	Lab system: RAMAN measurement channel, Sample / probe, Acquisition + reference data → Spectra + reference values	Sensor system: RAMAN sensor, Process probe(s), Acquisition + control interface → Process monitoring
2. UV/Vis (comparison)	Models: glucose concentration calibration, viable biomass calibration → Process predictions	Lab system: UVVIS measurement channel, Sample / probe, Acquisition + reference data → Spectra + reference values	Sensor system: UVVIS sensor, Process probe(s), Acquisition + control interface → Process monitoring
3. Fluorescence (conditional)	Models: glucose concentration calibration, viable biomass calibration → Process predictions	Lab system: FLUO measurement channel, Sample / probe, Acquisition + reference data → Spectra + reference values	Sensor system: FLUO sensor, Process probe(s), Acquisition + control interface → Process monitoring

Council

Winner: draft B; agreement 67% (2/3 drafts share the lead method(s) RAMAN; 3/3 share the status 'provisional_assessment')

- Draft A (openai/gpt-oss-120b): judge score 4.5, gate failures: G1 Sourced claims verified on the cited page, G2 No unsupported numbers
- Draft B (qwen/qwen3.8-27b): judge score 5.0, gate failures: G1 Sourced claims verified on the cited page, G2 No unsupported numbers
- Draft C (openai/gpt-oss-20b): judge score 3.75, gate failures: G1 Sourced claims verified on the cited page, G2 No unsupported numbers

Generated by the AFS assessment assistant (run 12bd4ae4dd, mode council, server default models; models used: openai/gpt-oss-120b, openai/gpt-oss-20b, openai/gpt-oss-20b -> rule fallback, qwen/qwen3.8-27b).

h4 — First technical assessment

> **Enquiry:** We want to investigate water content in a petroleum line using MIR ATR. The analyzer would be in another building. We already have a silica fiber cable from a visible or NIR measurement setup and would like to reuse it. The link length, routing and operating conditions have not yet been supplied. Can the assistant approve this concept?
>
> **Answers added:** Path C - Variables per sensor: One

Quality checks: 15/16 · **Sourced claims verified:** 7/10 · **Unsupported numbers:** 0

01. Decision Status

Provisional assessment · **Confidence:** insufficient (computed, not self-reported)

The concept of using MIR ATR for water content in petroleum is supported by lab data, but the reuse of the existing silica fiber is not possible; a new MIR-compatible fiber is required.

Problems found in the request

- Silica fiber cannot carry MIR: Silica fibers transmit only up to about 2.4 um (VIS-NIR silica 350-2400 nm). MIR ATR needs chalcogenide (CIR, about 1.1-6.5 um) or polycrystalline (PIR, about 3-17 um) fiber, so an existing silica link cannot be approved for MIR. [S02 p.6, S02 p.9]

The customer's own claims, tested first

- **partly supported**: "MIR ATR can measure water content in petroleum". Lab spectra show absorbance increases with water content between 4000–3000 cm-1, but no process installation, accuracy or detection limit is provided. [S11 p.13]
- **contradicted**: "reuse existing silica fiber cable". Silica fibers transmit only up to ~2.4 µm; MIR ATR requires chalcogenide or polycrystalline fiber, so a silica link cannot be approved for MIR. [S02 p.6] [S02 p.9]
- **contradicted**: "Can the assistant approve this concept?". Given the fiber incompatibility and lack of process-level evidence for MIR ATR water measurement, the concept cannot be approved. [S02 p.6] [S11 p.13]

To raise confidence, answer: Is the sample clear, turbid or opaque, and does this change during the process? / What concentration range must be covered, and what is the smallest change that matters? / Which independent reference method (and its uncertainty) can provide calibration values?

02. Requirements stated

- Water content in petroleum line
- MIR ATR method
- Remote analyzer in another building
- Reuse existing silica fiber cable

03. Key unknowns

- Link length, routing, and operating conditions (temperature, pressure, flow) are not supplied.
- The exact petroleum matrix and water content range are not specified.
- The client's acceptance criteria for accuracy and precision are not defined.

Assumptions made

- Assumption: The client's 'silica fiber cable' is a standard VIS-NIR silica fiber, not a specialized MIR fiber.
- Assumption: The client's 'petroleum line' implies a liquid hydrocarbon matrix, consistent with the AFS lab application.

- Assumption: The client's request for 'Path C' (one variable per sensor) is a preference for a simplified sensor architecture, but the primary technical blocker is the fiber type, not the sensor count.

04. Clarification questions

1. **What is the exact chemical composition of the petroleum stream (e.g., crude, refined diesel, gasoline) and the expected water content range?** *(high impact)*. Why: The AFS lab data covers 0.1-2.0% water in oil; if the client's range is different or the matrix is complex, the calibration strategy and method suitability change significantly.
2. **What are the specific acceptance criteria for water content measurement (e.g., accuracy, precision, response time) and the required operating temperature/pressure of the line?** *(high impact)*. Why: MIR ATR performance is highly dependent on temperature and pressure; without these, a feasibility plan cannot be designed, and the 'remote' aspect (link length) cannot be validated.
3. **Is the existing silica fiber cable the only asset to be reused, or is the client open to replacing it with a MIR-compatible fiber (chalcogenide or polycrystalline)?** *(medium impact)*. Why: The silica fiber is physically incompatible with MIR. Knowing if replacement is an option determines if the project can proceed with MIR or if NIR/Raman must be prioritized.
4. **What is the maximum acceptable link length between the analyzer and the process line?** *(medium impact)*. Why: MIR fiber transmission is limited compared to NIR; the link length dictates the fiber type (CIR vs PIR) and the feasibility of the remote setup.
5. **Are there any safety or regulatory constraints (e.g., ATEX, hazardous area) for the probe installation in the petroleum line?** *(low impact)*. Why: This affects the probe design (PEEK shaft, sealing) and the choice of materials, which must be compatible with the petroleum matrix.

05. Recommendation

MIR ATR (primary)

- AFS has lab data for MIR ATR measurement of water content in petroleum, showing absorbance increases with water content between 4000 and 3000 cm-1. [S02 p.13] *(sourced)*
- The existing silica fiber cable cannot be reused for MIR ATR because silica fibers transmit only up to about 2.4 um, while MIR ATR requires wavelengths above 2.5 um. [S02 p.6] *(sourced ⚠ unverified)*
- A new MIR-compatible fiber (chalcogenide or polycrystalline) is required for the remote analyzer setup. *(proposal)*
- *Applies if:* Only if the client agrees to replace the silica fiber with a MIR-compatible fiber and provides representative samples with reference water content.

NIR (diffuse reflection) (comparison)

- NIR can use the existing silica fiber cable and is suitable for remote measurements. [S01 p.5] *(sourced)*
- NIR moisture bands are a comparison option for water content in petroleum. [S02 p.13] *(sourced)*
- A chemometric model must be built and validated on the client's petroleum matrix. *(proposal)*
- *Applies if:* Only if the client wants to reuse the existing silica fiber and the NIR model meets the acceptance criteria.

Method screening (deterministic, from sourced rules)

Method	Score	Verdict
MIR spectroscopy (ATR)	4	lead
NIR spectroscopy	1	candidate
Raman spectroscopy	0	weak
UV/Vis spectroscopy	0	weak
Fluorescence spectroscopy	0	weak

06. Measurement concept

- AFS has lab data for MIR ATR measurement of water content in petroleum, showing absorbance increases with water content between 4000 and 3000 cm-1. [S02 p.13] *(sourced)*
- The existing silica fiber cable cannot be reused for MIR ATR because silica fibers transmit only up to about 2.4 um, while MIR ATR requires wavelengths above 2.5 um. [S02 p.6] *(sourced ⚠ unverified)*
- A new MIR-compatible fiber (chalcogenide or polycrystalline) is required for the remote analyzer setup. [S02 p.9] *(sourced ⚠ unverified)*
- The final method selection depends on the specific matrix, operating conditions, and acceptance criteria, which are not yet defined. [S01 p.5] *(sourced)*

07. Spectral range

- MIR ATR: 4000-3000 cm-1 (water absorbance band) [S02 p.13] *(sourced)*
- NIR: 700-2500 nm (moisture bands) [S01 p.5] *(sourced)*

08. Risks and uncertainty

- The client's petroleum matrix may have different spectral properties than the AFS lab data, requiring a new calibration model. *(assumption)*
- The operating conditions (temperature, pressure, flow) may affect the MIR ATR signal and the fiber performance. *(assumption)*
- The link length and routing may introduce additional losses or reflections, affecting the signal-to-noise ratio. *(assumption)*
- The client's acceptance criteria for accuracy and precision may not be met by the MIR ATR method, requiring a different method or a combined model. *(assumption)*

Confidence: insufficient

- Information: low (2/10 decision-relevant facts known; target defined: yes)
- Evidence: high (Direct application in AFS sources: R_WATER_IN_HYDROCARBON)
- Scope fit: high (Requested use matches the scope of the cited evidence)
- Client premises: low (2 contradicted, 0 unverifiable, 1 supported)
- Verification: medium (7/10 sourced claims verified; 0 unsupported numbers)
- Model agreement: low (1/3 drafts share the lead method(s) none; 3/3 share the status 'provisional_assessment' (2 model families))
- Cap: R_FIBER_SILICA_MIR: cap low
- Cap: R_CERTAINTY_DEMAND: cap low
- Cap: Council disagreement: lowered one level

09. Decision basis

- Calibration $R^2 > 0.95$ and prediction error < client-agreed limit

10. Sources

- **S01 p.5**: Multispectral Laboratory of Optical Analysis Methods: purpose, methods, and applications (art photonics / MIR Photonics) *(overview)*
  - "NIR spectroscopy, with its relatively large penetration depth (up to several millimetres) even into dense solid samples due to low absorption and strong scattering, allows the use of ordinary glass for optics"
  - "selecting a spectral method for a specific analytical problem is a complex task requiring specialized knowledge in chemistry, physics, and mathematics."
- **S02 p.6**: Process-Control in-line with Spectral Fiber Systems (AFS slides) *(company slides)*
  - "Silica fibers transmit only up to about 2.4 um" ⚠ unverified
  - "Silica fibers transmit only up to about 2.4 um (VIS-NIR silica 350-2400 nm). MIR ATR needs chalcogenide (CIR, about 1.1-6.5 um) or polycrystalline (PIR, about 3-17 um) fiber, so an existing silica link cannot be approved for MIR."
- **S02 p.9**: Process-Control in-line with Spectral Fiber Systems (AFS slides) *(company slides)*
  - "MIR ATR needs chalcogenide (CIR, about 1.1-6.5 um) or polycrystalline (PIR, about 3-17 um) fiber" ⚠ unverified
  - "Silica fibers transmit only up to about 2.4 um (VIS-NIR silica 350-2400 nm). MIR ATR needs chalcogenide (CIR, about 1.1-6.5 um) or polycrystalline (PIR, about 3-17 um) fiber, so an existing silica link cannot be approved for MIR."
- **S02 p.13**: Process-Control in-line with Spectral Fiber Systems (AFS slides) *(company slides)*
  - "Application: measurement of water content in petroleum"
  - "MIR absorbance spectra from 4000 to 2000 cm-1 of pure oil and of oil with 0.1%, 0.5%, 1.0% and 2.0% water; absorbance between about 4000 and 3000 cm-1 increases with water content"
- **S11 p.13**: S11
  - "Application: measurement of water content in petroleum"

Verifier warnings

- recommendations[0].rationale[1]: unverified S02 p.6: quote not on page (52)
- measurement_concept[1]: unverified S02 p.6: quote not on page (52)
- measurement_concept[2]: unverified S02 p.9: quote not on page (46)

11. First feasibility experiment

Samples

- Representative petroleum samples with 0.1–2.0 % water, spiked and unspiked
- Process samples at operating temperature and pressure **Reference measurements**
- Independent reference measurements (e.g., Karl Fischer titration) for each sample **Comparison**
- NIR diffuse reflection vs. MIR ATR (if MIR fiber link installed) **Steps**

1. Collect and prepare representative samples.
2. Acquire NIR spectra with existing silica fiber and build chemometric model.
3. If MIR fiber link is available, acquire MIR ATR spectra and build separate model.
4. Validate models against reference data.

5. Compare performance and decide on deployment.

Next step with AFS

- Multi-method lab trial comparing MIR ATR (with new fiber) and NIR (with existing fiber) on representative petroleum samples.
- Likely path: A - Path A turns spectral data into real-time concentration values for process automation; a central full-spectrum analyzer gives the higher accuracy.
- Path A (chosen by the expert): **fits**. Path A turns spectral data into real-time concentration values for process automation; a central full-spectrum analyzer gives the higher accuracy. [S02 p.20, S02 p.18]
- Path B (chosen by the expert): **open**. No finding speaks for or against this path yet; the lab trial decides. [S02 p.20]
- Path C (chosen by the expert): **open**. No finding speaks for or against this path yet; the lab trial decides. [S02 p.20]

Proposed setup (ranked by recommendation; validate in a lab trial)

System	Path A: Chemometric model for process automation	Path B: Multi-spectral system for your R&D lab	Path C: Low-cost spectral sensors
1. MIR ATR (primary)	Models: water content calibration → Process predictions	Lab system: MIR measurement channel, Sample / probe, Acquisition + reference data → Spectra + reference values	Sensor system: MIR sensor, Process probe(s), Acquisition + control interface → Process monitoring
2. NIR (diffuse reflection) (comparison)	Models: water content calibration → Process predictions	Lab system: NIR measurement channel, Sample / probe, Acquisition + reference data → Spectra + reference values	Sensor system: NIR sensor, Process probe(s), Acquisition + control interface → Process monitoring

Council

Winner: draft B; agreement 33% (1/3 drafts share the lead method(s) none; 3/3 share the status 'provisional_assessment')

- Draft A (openai/gpt-oss-120b): judge score 4.75, gate failures: G1 Sourced claims verified on the cited page
- Draft B (qwen/qwen3.8-27b): judge score 4.92, gate failures: none
- Draft C (openai/gpt-oss-20b): judge score 4.17, gate failures: none

Generated by the AFS assessment assistant (run e897dd847e, mode council, server default models; models used: openai/gpt-oss-120b, openai/gpt-oss-20b, qwen/qwen3.8-27b).

h5 — First technical assessment

> **Enquiry:** We have a liquid process that occasionally goes off specification. We suspect contamination, but we cannot yet name the contaminant or the affected quality variable. We would like you to recommend the best spectroscopy method and exact sensor configuration immediately. What information and first experiment should come before that decision?

Quality checks: 13/16 · **Sourced claims verified:** 3/3 · **Unsupported numbers:** 3

01. Decision Status

Provisional assessment · **Confidence:** insufficient (computed, not self-reported)

An exact method and sensor configuration cannot be finalized until the contaminant identity, its optical signature, and the quality variable to monitor are defined; we recommend Raman spectroscopy as the most promising starting point and outline the information and first experiment needed.

The customer's own claims, tested first

- **unverifiable**: "They want an exact recommendation of method and sensor configuration immediately.". An exact configuration cannot be justified before the target, sample optics, conditions and acceptance criteria are known; method selection depends on many factors and is decided by comparison on real samples. [S01 p.5] [S02 p.3]
- **unverifiable**: "They ask what information and first experiment should come before that decision.". The supplied material does not settle this; it needs to be tested.

To raise confidence, answer: Is the sample clear, turbid or opaque, and does this change during the process? / What concentration range must be covered, and what is the smallest change that matters? / Which independent reference method (and its uncertainty) can provide calibration values?

02. Requirements stated

- Recommend best spectroscopy method for a liquid process with unknown contaminant
- Specify sensor configuration
- Identify prerequisite information and first experiment

03. Key unknowns

- Identity of the contaminant
- Optical behavior of the contaminant in the aqueous matrix
- Target quality variable to monitor
- Typical concentration level of the contaminant
- Matrix composition and variability

Assumptions made

- The contaminant is chemically distinct enough to produce a Raman-active vibration or MIR absorption band.
- Process temperature and pressure are within the operating limits of standard Raman probes.
- Sufficient sample volume can be diverted for offline reference analysis.

04. Clarification questions

1. **What are the most critical quality attributes (e.g., purity, viscosity, color) that could be impacted by the suspected contamination?** *(high impact)*. Why: Knowing the quality variable defines the required sensitivity, selectivity, and whether a quantitative or classification model is needed.
2. **Do you have any prior analytical data (e.g., GC-MS, HPLC) that hint at possible contaminant classes or functional groups?** *(high impact)*. Why: Even coarse chemical class information narrows the spectral region and probe type (Raman vs. MIR).
3. **What is the typical optical density or turbidity of the process liquid under normal operation?** *(medium impact)*. Why: Optical density determines whether transmission, ATR, or back-scatter geometries are feasible.
4. **Are there constraints on probe installation (e.g., temperature, pressure, cleaning cycles) at the measurement point?** *(medium impact)*. Why: Installation constraints affect the choice between a local sensor (path C) and a central analyzer with fiber links.
5. **What acceptance criteria (e.g., false-negative rate, detection limit) would satisfy process control?** *(low impact)*. Why: Criteria drive the required detection limit and thus influence method suitability.

05. Recommendation

Raman (90° back-scatter probe, liquid immersion) (primary)

- Raman is compatible with aqueous liquids and provides high selectivity for many organic contaminants. *(assumption)*
- A probe can be installed locally at the process point, matching the suggested Path C for lower installation cost. *(proposal)*
- *Applies if:* Only if a representative set of off-spec and in-spec liquid samples can be collected for model building.

MIR-ATR (chalcogenide fiber, <50 µm path) (conditional)

- MIR provides well-defined absorption peaks that can aid identification if the contaminant has strong IR bands. [S01 p.5] *(sourced)*
- Requires a specialty fiber; feasibility should be tested alongside Raman. *(proposal)*
- *Applies if:* If Raman sensitivity proves insufficient after initial trials.

NIR (transmission cuvette) (not recommended)

- NIR sensitivity at trace levels is limited for unknown contaminants. *(assumption)*

Method screening (deterministic, from sourced rules)

Method	Score	Verdict
Raman spectroscopy	1	candidate
MIR spectroscopy (ATR)	0	weak
UV/Vis spectroscopy	0	weak
Fluorescence spectroscopy	0	weak
NIR spectroscopy	-1	weak

06. Measurement concept

- Raman spectroscopy with a 90° back-scatter probe is well-suited for aqueous liquids because water exhibits low Raman absorbance, providing high signal-to-noise for dissolved species. [S01 p.5] *(sourced)*
- If the contaminant has strong mid-IR absorption, an ATR probe could capture well-defined peaks, but requires a chalcogenide or polycrystalline fiber for MIR transmission. [S01 p.5] *(sourced)*
- NIR is less attractive for trace-level detection because sensitivity drops below ~0.1 % for unknown species. *(assumption)*

07. Spectral range

- Raman: excitation typically 785 nm or 532 nm; Raman shift measured in cm^{-1} covering 200–3500 cm^{-1}. *(proposal)*
- MIR-ATR: 4000–650 cm^{-1} (2500–15000 nm) using chalcogenide fiber. *(proposal)*

08. Risks and uncertainty

- If the contaminant does not have a strong Raman-active vibration, detection may be limited. *(assumption)*
- ATR probe fouling in a process stream could degrade signal over time. *(assumption)*
- Raman signal can be weak for low-concentration species; may require longer integration or higher laser power, raising safety considerations. *(assumption)*

Confidence: insufficient

- Information: insufficient (1/10 decision-relevant facts known; target defined: partial)
- Evidence: low (Only general method knowledge applies; no direct application example in the sources)
- Scope fit: high (Requested use matches the scope of the cited evidence)
- Client premises: medium (0 contradicted, 2 unverifiable, 0 supported)
- Verification: high (3/3 sourced claims verified; 3 unsupported numbers)
- Model agreement: low (1/2 drafts share the lead method(s) RAMAN; 1/2 share the status 'provisional_assessment' (2 model families))
- Cap: R_CERTAINTY_DEMAND: cap low

09. Decision basis

- Ability to detect the contaminant at the concentration observed in off-spec samples (client-agreed detection limit)
- Robustness of the probe under process temperature/pressure conditions
- Ease of integration at the measurement point (cost, cleaning, maintenance)

10. Sources

- **S01 p.5**: Multispectral Laboratory of Optical Analysis Methods: purpose, methods, and applications (art photonics / MIR Photonics) *(overview)*
  - "strong absorption in the mid-IR region limits the corresponding spectral method to the analysis of very thin samples (usually less than 50 µm), but it opens up the possibility of using attenuated total reflectance (ATR)"
  - "the spectra of vibrational transitions in Raman and mid-IR spectroscopy carry complementary qualitative and quantitative information about the sample's composition."

  - "NIR spectroscopy, with its relatively large penetration depth (up to several millimetres) even into dense solid samples due to low absorption and strong scattering, allows the use of ordinary glass for optics"
- **S02 p.3**: Process-Control in-line with Spectral Fiber Systems (AFS slides) *(company slides)*
  - "Multi-Spectral Fiber Systems allow to test 4 spectroscopy methods in 0.3-16µm range"

11. First feasibility experiment

Samples

- In-spec liquid from normal operation
- Off-spec liquid collected during a deviation
- Spiked laboratory samples covering a range of plausible contaminants (e.g., organic solvents, salts, oils) **Reference measurements**
- Independent laboratory analysis (e.g., GC-MS or HPLC) to confirm presence/absence of contaminants in each sample **Comparison**
- Raman probe vs. MIR-ATR probe (if available) on identical samples
- Signal-to-noise and detection limit assessment for each method **Steps**

1. Collect representative liquid samples (in-spec, off-spec, and spiked) and perform reference analysis.
2. Acquire Raman spectra with a standard 90° back-scatter probe (785 nm laser) on all samples.
3. If possible, acquire MIR-ATR spectra using a short-path ATR probe on the same set.
4. Build preliminary chemometric classification models (e.g., PLS-DA) for each method using the reference labels.
5. Evaluate detection limits, repeatability, and probe durability against client-agreed criteria.
6. Decide on the final sensor configuration (Raman only or Raman + MIR) and proceed to pilot-scale installation.

Next step with AFS

- Perform a lab-scale comparative trial of Raman and optional MIR-ATR probes on representative process liquids to determine the most sensitive and robust configuration for on-site sensor deployment.
- Likely path: C - Path C favours low-cost local sensors; the trial will confirm which local probe meets the required detection performance.
- Path A (chosen by the expert): **open**. No finding speaks for or against this path yet; the lab trial decides. [S02 p.20]
- Path B (chosen by the expert): **open**. No finding speaks for or against this path yet; the lab trial decides. [S02 p.20]
- Path C (chosen by the expert): **fits**. Sensors installed at the measuring points give lower installation cost than a central analyzer with long fibers. [S02 p.18, S02 p.19]

Proposed setup (ranked by recommendation; validate in a lab trial)

System	Path A: Chemometric model for process automation	Path B: Multi-spectral system for your R&D lab	Path C: Low-cost spectral sensors
1. Raman (90° back-scatter probe, liquid immersion) (primary)	Models: Target calibration → Process predictions	Lab system: RAMAN measurement channel, Sample / probe, Acquisition + reference data → Spectra + reference values	Sensor system: RAMAN sensor, Process probe(s), Acquisition + control interface → Process monitoring

System	Path A: Chemometric model for process automation	Path B: Multi-spectral system for your R&D lab	Path C: Low-cost spectral sensors
2. MIR-ATR (chalcogenide fiber, <50 µm path) (conditional)	Models: Target calibration → Process predictions	Lab system: MIR measurement channel, Sample / probe, Acquisition + reference data → Spectra + reference values	Sensor system: MIR sensor, Process probe(s), Acquisition + control interface → Process monitoring

Council

Winner: draft A; agreement 50% (1/2 drafts share the lead method(s) RAMAN; 1/2 share the status 'provisional_assessment')

- Draft A (openai/gpt-oss-120b): judge score 4.67, gate failures: G1 Sourced claims verified on the cited page, G2 No unsupported numbers
- Draft B (qwen/qwen3.8-27b): judge score 0.0, gate failures: none
- Draft C (openai/gpt-oss-20b): judge score 4.33, gate failures: G1 Sourced claims verified on the cited page

Generated by the AFS assessment assistant (run 9d2f822bd5, mode council, server default models; models used: openai/gpt-oss-120b, openai/gpt-oss-20b, openai/gpt-oss-20b -> rule fallback, qwen/qwen3.8-27b).

h6 — First technical assessment

> **Enquiry:** We read the supplied LED NIR kidney study and want to use the same concept to guide tumor boundary decisions during live surgery. Can we quote its findings as proof that the proposed sensor is ready for that use? We have no new in vivo data and no validation on our own setup. What assessment and next feasibility steps are appropriate?

Quality checks: 12/16 · **Sourced claims verified:** 5/8 · **Unsupported numbers:** 0

01. Decision Status

Outside the supported scope · **Confidence:** insufficient (computed, not self-reported)

The LED NIR kidney study is an in vitro proof of concept and cannot be quoted as proof that the sensor is ready for live surgical decision-making; clinical validation and regulatory work are required.

Problems found in the request

- Clinical or in vivo use is outside the supported evidence: The supplied clinical evidence is an in vitro proof of concept (two LED NIR prototypes, 14 in vitro samples from 7 patients) and model-tissue lab work. It cannot be quoted as proof that a sensor is ready to guide decisions in live surgery; that use is outside the supported scope and would need clinical validation and regulatory work beyond this assessment. [S03 p.1, S03 p.14, S04 p.1]

The customer's own claims, tested first

- **contradicted**: "LED NIR kidney study demonstrates sensor readiness for tumor boundary decisions during live surgery.". The study is an in vitro proof-of-concept with only 14 samples and no in-vivo data; it cannot be quoted as proof that the sensor is ready for live surgical decision-making. [S03 p.1] [S03 p.14]
- **contradicted**: "Can we quote its findings as proof that the proposed sensor is ready for that use?". Rule R_CLINICAL_SCOPE states that the in vitro evidence cannot be used as proof of readiness for live surgery; regulatory and clinical validation are required. [S03 p.1] [S03 p.14]
- **partly supported**: "What assessment and next feasibility steps are appropriate?". The evidence supports that further validation is needed: improve LED number/wavelength, optimize measurement conditions, and perform in-vivo testing with real surgical samples before clinical use. [S03 p.13] [S03 p.13]

To raise confidence, answer: What is the sample (liquid, powder, melt, tissue...) and its main components? / Is the sample clear, turbid or opaque, and does this change during the process? / What concentration range must be covered, and what is the smallest change that matters?

02. Requirements stated

- Client wants to use LED NIR kidney study to justify sensor for tumor boundary during live surgery.
- Client lacks new in vivo data or validation on their setup.
- Client asks if findings can be quoted as proof and what assessment/feasibility steps are needed.

03. Key unknowns

- Whether the study’s findings are directly applicable to the specific surgical environment.
- Specific sensor hardware and calibration used in the study.
- Patient population and tumor types in the study.

04. Clarification questions

1. **What is the specific clinical endpoint and acceptance criteria for 'tumor boundary decisions' (e.g., margin clearance, real-time classification accuracy) that the sensor must meet?** *(high impact)*. Why: The current study is an in vitro proof of concept; defining the clinical target is required to determine the scope of necessary validation and regulatory pathway.
2. **Do you have access to a representative set of ex vivo tissue samples (healthy and tumor) from the relevant organ and tumor types to build a validation dataset?** *(high impact)*. Why: The study notes limited sample availability (14 samples from 7 patients); a robust clinical validation requires a larger, representative dataset to avoid overfitting and ensure generalizability.
3. **What are the specific hardware constraints and measurement conditions (distance, angle, light intensity) in your intended surgical environment?** *(medium impact)*. Why: The study identifies optimization of measurement conditions as a key area for improvement; understanding the operational context is necessary to assess feasibility and required sensor modifications.
4. **Is there a plan for regulatory engagement (e.g., FDA, CE marking) and clinical trial design to support the transition from proof of concept to clinical use?** *(medium impact)*. Why: Clinical decision support requires regulatory approval; understanding the current stage of regulatory planning helps in structuring the next feasibility steps.
5. **Are there existing in vivo data or pilot studies from other groups that could be used to inform the sensor design or validation strategy?** *(low impact)*. Why: While the current study is in vitro, leveraging existing in vivo data could help refine the sensor design and reduce the risk of failure in later clinical trials.

05. Recommendation

NIR (LED-based sensor) (primary)

- The study demonstrates the feasibility of an LED-based NIR sensor for tumor detection in an in vitro setting. [S03 p.1] *(sourced)*
- The study identifies the need to improve the accuracy and specificity of detection for clinical use. [S03 p.14] *(sourced ⚠ scope warning)*
- Optimize the LED number, wavelength, and measurement conditions to reduce measurement variance and improve sensor performance. *(proposal)*
- *Applies if:* Only if a representative set of ex vivo tissue samples and a defined clinical endpoint are available.

Method screening (deterministic, from sourced rules)

Method	Score	Verdict
NIR spectroscopy	0	weak
MIR spectroscopy (ATR)	0	weak
Raman spectroscopy	0	weak
UV/Vis spectroscopy	0	weak
Fluorescence spectroscopy	0	weak

06. Measurement concept

- The study used two LED-based NIR sensor prototypes with 14 in vitro clinical samples from 7 patients, showing high measurement variance in cancer tissue samples. [S03 p.1] [S03 p.14] *(sourced ⚠ scope warning)*

- The study identifies the need to improve the accuracy and specificity of detection, as false negatives are unacceptable in oncological diagnostics. [S03 p.14] *(sourced)*
- Optimize the LED number, wavelength, and measurement conditions (distance, angle, light intensity) to reduce measurement variance and improve sensor performance. *(proposal)*

07. Spectral range

- NIR light penetrates into the sample by up to 3 mm, which is advantageous for tissue diagnostics. [S03 p.2] *(sourced)*
- The study used four LEDs with emission wavelengths in the NIR range. [S03 p.14] *(sourced)*

08. Risks and uncertainty

- The current sensor prototype has high measurement variance in cancer tissue samples, which could lead to false negatives in clinical use. [S03 p.14] *(sourced ⚠ scope warning)*
- The study is based on a very limited number of patients (7), which may not be representative of the broader patient population. [S03 p.6] *(sourced)*
- The sensor's performance in a live surgical environment may differ from the in vitro conditions due to factors such as tissue movement, blood presence, and environmental effects. *(assumption)*
- Regulatory approval for clinical use will require extensive clinical validation and may take several years. *(assumption)*

Confidence: insufficient

- Information: insufficient (0/10 decision-relevant facts known; target defined: partial)
- Evidence: insufficient (Only general method knowledge applies; no direct application example in the sources)
- Scope fit: insufficient (Requested use (clinical/in vivo) exceeds the in-vitro / lab scope of the evidence)
- Client premises: low (2 contradicted, 0 unverifiable, 1 supported)
- Verification: low (5/8 sourced claims verified; 0 unsupported numbers)
- Model agreement: low (1/2 drafts share the lead method(s) none; 2/2 share the status 'outside_supported_scope' (2 model families))
- Cap: R_CLINICAL_SCOPE: cap insufficient
- Cap: R_CERTAINTY_DEMAND: cap low

09. Decision basis

- Client-agreed acceptance criteria for tumor boundary classification accuracy and specificity on independent ex vivo tissue samples.
- Demonstration of reduced measurement variance compared to the current prototype.
- Successful optimization of LED number, wavelength, and measurement conditions to improve sensor performance.

10. Sources

- **S03 p.1**: Development and Testing of an LED-Based Near-Infrared Sensor for Human Kidney Tumor Diagnostics (Sensors 2017) *(proof_of_concept_in_vitro)*
  - "Two sensor prototypes were tested using 14 in vitro clinical samples of 7 different patients."
  - "the present prototype should be further improved, the results reported here can be considered as a successful feasibility study of an LED-based NIR sensor for cancer diagnostics aimed at in vivo applications."

- **S03 p.2**: Development and Testing of an LED-Based Near-Infrared Sensor for Human Kidney Tumor Diagnostics (Sensors 2017) *(proof_of_concept_in_vitro)*
  - "It is important that the NIR light penetrates into the sample by up to 3 mm [20,21]; this depth cannot be reached by suitable spectroscopic techniques in other wavelength regions."
- **S03 p.6**: Development and Testing of an LED-Based Near-Infrared Sensor for Human Kidney Tumor Diagnostics (Sensors 2017) *(proof_of_concept_in_vitro)*
  - "conclusions about the general suitability of the NIR sensor analysis for cancer diagnostics and, in particular, of four theoretically chosen LEDs are made on the basis of a very limited number of investigated patients."
- **S03 p.13**: Development and Testing of an LED-Based Near-Infrared Sensor for Human Kidney Tumor Diagnostics (Sensors 2017) *(proof_of_concept_in_vitro)*
  - "The main requirement for the sensor improvement is therefore to increase the accuracy and specificity of detection, as FN are unacceptable outcomes in oncological diagnostics."
  - "The directions for the further improvement of the sensor are (1) LED number and wavelength optimization, (2) optimization of the measurement conditions (distance, angle, light intensity, spectral reference, probe construction, elimination of environmental effects, etc.), and (3) data processing optimization."
- **S03 p.14**: Development and Testing of an LED-Based Near-Infrared Sensor for Human Kidney Tumor Diagnostics (Sensors 2017) *(proof_of_concept_in_vitro)*
  - "The main requirement for the sensor improvement is therefore to increase the accuracy and specificity of detection, as FN are unacceptable outcomes in oncological diagnostics." ⚠ scope_warning
  - "The main disadvantage of the LED-based sensor in its present form is a very high measurement variance in cancer tissue samples (Figure 9)." ⚠ scope_warning
  - "Only four LEDs with emission wavelengths in the NIR range were employed."
  - "The sensor’s general sensitivity to the kidney tumors, illustrated here, provides the necessary motivation for further development of the method."

Verifier warnings

- recommendations[0].rationale[1]: scope_warning S03 is in-vitro proof of concept; it cannot support a readiness claim
- measurement_concept[0]: scope_warning S03 is in-vitro proof of concept; it cannot support a readiness claim
- risks[0]: scope_warning S03 is in-vitro proof of concept; it cannot support a readiness claim

11. First feasibility experiment

Samples

- Representative ex vivo tissue samples (healthy and tumor) from the relevant organ and tumor types, including physical variation (e.g., tissue density, moisture content). **Reference measurements**
- Independent reference measurements using histopathology (gold standard) on the same ex vivo tissue samples.
- Comparison with full-range NIR spectroscopy results on the same samples to validate the LED-based approach. **Comparison**
- Compare the performance of the current LED-based sensor against a full-range NIR spectrometer on the same ex vivo tissue samples.
- Evaluate the impact of varying the number of LEDs and their wavelengths on the sensor's accuracy and specificity.

- Assess the robustness of the sensor under different measurement conditions (distance, angle, light intensity) to simulate real-world surgical variability. **Steps**

1. Define the clinical endpoint and acceptance criteria for tumor boundary decisions.
2. Collect a representative set of ex vivo tissue samples (healthy and tumor) from the relevant organ and tumor types.
3. Optimize the LED number, wavelength, and measurement conditions (distance, angle, light intensity) to improve sensor accuracy and specificity.
4. Validate the optimized sensor on the ex vivo tissue samples using histopathology as the reference standard.
5. Assess the robustness of the sensor under different measurement conditions to simulate real-world surgical variability.
6. Prepare a regulatory submission plan for clinical validation and regulatory approval.

Next step with AFS

- Define the clinical endpoint and acceptance criteria, and collect a representative set of ex vivo tissue samples for sensor optimization and validation.
- Likely path: undecided - Too early: the target or the scope must be settled first.
- Path A (chosen by the expert): **premature**. Clinical decision support needs clinical validation before any routine model. [S03 p.1, S03 p.14]
- Path B (chosen by the expert): **premature**. A lab system can support further ex vivo research, but not the requested clinical use. [S03 p.1, S03 p.14]
- Path C (chosen by the expert): **premature**. The LED sensor is an in vitro proof of concept; a sensor for clinical decisions needs clinical validation first. [S03 p.1, S03 p.14]

If you proceed anyway with Path A (Path A (chemometric model for process automation) is premature here: Clinical decision support needs clinical validation before any routine model.)

1. Collect representative samples across the client-agreed range, each with an independent reference value.
2. Record full spectra with the short-listed method(s) and build a calibration model; validate it on held-out batches or runs.
3. Agree how the model is maintained (recalibration triggers, matrix changes) before it is used for process control. *Sources: S01 p.5, S02 p.20*

If you proceed anyway with Path B (Path B (customised multi-spectral system for R&D) is premature here: A lab system can support further ex vivo research, but not the requested clinical use.)

1. Specify the lab kit from the methods short-listed in this assessment: spectrometers, fiber probes, software.
2. Run the method comparison on the customer's own samples in their lab.
3. Use the results as the basis for a later process solution (Path A or C). *Sources: S02 p.20, S01 p.4*

If you proceed anyway with Path C (Path C (customised spectral IoT sensor) is premature here: The LED sensor is an in vitro proof of concept; a sensor for clinical decisions needs clinical validation first.)

1. Run a lab trial with full-spectrum instruments on representative samples.
2. Select the most informative wavelengths and check that a few-band model reaches the client-agreed error on independent samples.
3. Build and test a prototype LED / filter sensor against the reference method before any volume order. *Sources: S01 p.9, S03 p.14, S02 p.18*

Proposed setup (ranked by recommendation; validate in a lab trial)

System	Path A: Chemometric model for process automation	Path B: Multi-spectral system for your R&D lab	Path C: Low-cost spectral sensors
1. NIR (LED-based sensor) (primary)	Models: Target calibration → Process predictions	Lab system: NIR measurement channel, Sample / probe, Acquisition + reference data → Spectra + reference values	Sensor system: NIR sensor, Process probe(s), Acquisition + control interface → Process monitoring

Council

Winner: draft B; agreement 50% (1/2 drafts share the lead method(s) none; 2/2 share the status 'outside_supported_scope')

- Draft A (openai/gpt-oss-120b): judge score 4.83, gate failures: G2 No unsupported numbers, G4 Evidence used within its scope
- Draft B (qwen/qwen3.8-27b): judge score 5.0, gate failures: G1 Sourced claims verified on the cited page, G4 Evidence used within its scope, H1 No guarantee or readiness wording
- Draft C (openai/gpt-oss-20b): judge score 0.0, gate failures: none

Generated by the AFS assessment assistant (run 36f004eb50, mode council, server default models; models used: openai/gpt-oss-120b, openai/gpt-oss-20b, qwen/qwen3.8-27b).